Spatial Reference

The **International Terrestrial Reference System** (**ITRS**) describes procedures for creating reference frames suitable for use with measurements on or near the Earth's surface. This is done in much the same way that a physical standard might be described as a set of procedures for creating a *realization* of that standard. The ITRS defines a geocentric system of coordinates using the SI system of measurement.

The **World Geodetic System** (**WGS**) is a standard for use in cartography, geodesy, and navigation including by GPS. It comprises a standard coordinate system for the Earth, a standard spheroidal reference surface (the *datum* or*reference ellipsoid*) for raw altitude data, and a gravitational equipotential surface (the *geoid*) that defines the *nominal sea level*.

The latest revision is **WGS 84** (aka **WGS 1984**, **EPSG:4326**), established in 1984 and last revised in 2004.[1] Earlier schemes included **WGS 72**, **WGS 66**, and **WGS 60**. WGS 84 is the reference coordinate system used by the Global Positioning System.

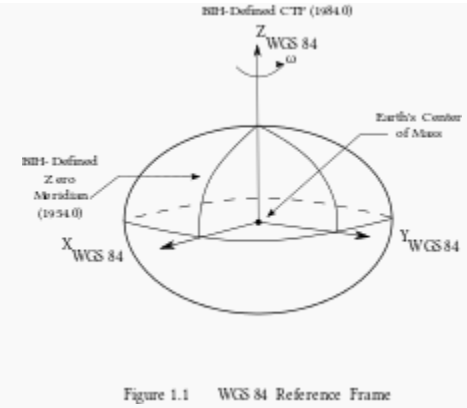
An **International Terrestrial Reference Frame** (**ITRF**) is a realization of the ITRS. New ITRF solutions are produced every few years, using the latest mathematical and surveying techniques to attempt to realize the ITRS as precisely as possible. Due to experimental error, any given ITRF will differ very slightly from any other realization of the ITRF. Also, the difference between the latest WGS84 and the latest ITRF is only a few centimeters.[1]

Practical navigation systems are in general referenced to a specific ITRF solution, or to their own coordinate systems which are then referenced to an ITRF solution.

The **Galileo Terrestrial Reference Frame** (**GTRF**) is used for the Galileo navigation system; currently defined as ITRF2005.

The ITRS and ITRF solutions are maintained by the International Earth Rotation and Reference Systems Service (IERS). GTRF is defined by the European Space Agency (ESA).

A new World Geodetic System: WGS 84[edit]



WGS 84 reference frame.

In the early 1980s the need for a new world geodetic system was generally recognized by the geodetic community and also within the US Department of Defense. WGS 72 no longer provided sufficient data, information, geographic coverage, or product accuracy for all then-current and anticipated applications. The means for producing a new WGS were available in the form of improved data, increased data coverage, new data types and improved techniques. GRS 80 parameters together with available Doppler, satellite laser ranging and Very Long Baseline Interferometry (VLBI) observations constituted significant new information. An outstanding new source of data had become available from satellite radar altimetry. Also available was an advanced least squares method called collocation which allowed for a consistent combination solution from different types of measurements all relative to the Earth's gravity field, i.e. geoid, gravity anomalies, deflections, dynamic Doppler, etc.

The new World Geodetic System was called WGS 84. It is currently the reference system being used by the Global Positioning System. It is geocentric and globally consistent within ±1 m. Current geodetic realizations of the geocentric reference system family International Terrestrial Reference

System (ITRS) maintained by the IERS are geocentric, and internally consistent, at the few-cm level, while still being metre-level consistent with WGS 84.

The WGS 84 originally used the GRS 80 reference ellipsoid, but has undergone some minor refinements in later editions since its initial publication. Most of these refinements are important for high-precision orbital calculations for satellites but have little practical effect on typical topographical uses. The following table lists the primary ellipsoid parameters.

Ellipsoid reference	Semi-major axis *a*	Semi-minor axis *b*	Inverse flattening (1/*f*)
GRS 80	6 378 137.0 m	≈ 6 356 752.314 140 m	298.257 222 100 882 711...
WGS 84	6 378 137.0 m	≈ 6 356 752.314 245 m	298.257 223 563

The very small difference in the flattening thus results in a tiny difference of 0.105 mm in the semi polar axis.